

Year 7 · Chapter 2 · Angles: homework answers (teacher copy)

Part 1: drill practice (question, lesson)

1. M (2.01)
2. N (2.01)
3. Y (2.01)
4. S (2.01)
5. FP and FD (2.01)
6. PK and PC (2.01)
7. FH and FK (2.01)
8. FS and FM (2.01)
9. 95° (2.02)
10. 55° (2.02)
11. 40° (2.02)
12. 120° (2.02)
13. obtuse (2.02)
14. acute (2.02)
15. reflex (2.02)
16. reflex (2.02)
17. obtuse (2.03)
18. reflex (2.03)
19. reflex (2.03)
20. acute (2.03)
21. 255° , reflex (2.03)
22. 255° , reflex (2.03)
23. 180° , straight (2.03)
24. 150° , obtuse (2.03)
25. 23° (2.04)
26. 62° (2.04)
27. 74° (2.04)
28. 89° (2.04)
29. C (2.04)
30. N (2.04)
31. C (2.04)
32. N (2.04)
33. $a = 26$ (2.05)
34. $a = 126$ (2.05)
35. $a = 36$ (2.05)
36. $a = 11$ (2.05)
37. $v = 117^\circ$, $n = 63^\circ$ (2.05)
38. $v = 50^\circ$, $n = 130^\circ$ (2.05)
39. $v = 19^\circ$, $n = 161^\circ$ (2.05)
40. $EF \perp GH$ (2.06)
41. $\parallel$ (2.06)
42. true (2.06)
43. false (2.06)
44. 45° (2.07)
45. 37° (2.07)
46. 79° (2.07)
47. 91° (2.07)
48. $a = 100$, $b = 80$ (2.07)
49. $a = 120$, $b = 60$ (2.07)
50. $a = 130$, $b = 50$ (2.07)
51. 48° (2.08)
52. 31° (2.08)

53. 41° (2.08)
54. 91° (2.08)
55. $a = 134^\circ$, $c = 46^\circ$ (2.08)
56. $a = 107^\circ$, $c = 73^\circ$ (2.08)
57. $a = 154^\circ$, $c = 26^\circ$ (2.08)
58. 33° (2.09)
59. 40° (2.09)
60. 35° (2.09)
61. 23° (2.09)
62. $y = 11$ (2.09)
63. $y = 19$ (2.09)
64. $y = 15$ (2.09)
65. corresponding (2.10)
66. corresponding (2.10)
67. 33° , 147° , 147° (2.10)
68. 93° , 87° , 87° (2.10)
69. 54° , 126° , 126° (2.10)
70. yes (2.11)
71. yes (2.11)
72. yes (2.11)
73. $x = 125$ (2.11)
74. $x = 105$ (2.11)
75. $x = 30$ (2.11)

Part 2: mixed revision (question, lesson)

76. M (Easy, 2.01)
77. 48° (Easy, 2.09)
78. True (Easy, 2.06)
79. obtuse (Easy, 2.03)
80. no (Easy, 2.11)
81. 37° (Easy, 2.08)
82. 120° (Easy, 2.05)
83. 135° (Easy, 2.02)
84. alternate, equal (Easy, 2.10)
85. 84° (Easy, 2.07)
86. 52° (Easy, 2.04)
87. $\angle CBE$ (Medium, 2.01)
88. 180° (Medium, 2.09)
89. e.g. wall and floor (Medium, 2.06)
90. reflex; straight (Medium, 2.03)
91. No: $105 + 85 = 190$, not 180 (Medium, 2.11)
92. Z (Medium, 2.08)
93. 38° (Medium, 2.05)
94. accurate 135° (Medium, 2.02)
95. $x = 130$, $y = 50$ (Medium, 2.10)
96. an F (Medium, 2.07)
97. 62° (Medium, 2.04)
98. two angles sharing arm BC (Challenging, 2.01)
99. 19 (201° to 219°) (Challenging, 2.03)
100. 115° , 65° , 115° (Challenging, 2.05)
101. $x = 20$ (Challenging, 2.07)
102. $y = 20$ (Challenging, 2.09)
103. e.g. corresponding angles 60° and 70° (Challenging, 2.11)